

# Jenny Vivian

📍 Australia, Austria, Italy

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## Research Interests

Tropical forest ecology, modification of the environment after deforestation, integrated approach, Machine Learning, eDNA

## Education

### University of the Sunshine Coast

Nov. 2023 – Aug. 2026

PhD in Forestry Studies

Thesis: “The effect of reforestation using *Acacia mangium* on above- and belowground ecosystem properties”

Principal Supervisor: Professor D. Lee; co-supervisors: Professor R. L. Chazdon, Professor A. Shapcott, Dr. A. A. Catling

### University of Parma

Sept. 2021 – Jul. 2023

Master’s Degree in Ecology and Ethology for nature conservation

Thesis: “The Colombian socio-ecological system: dynamics of deforestation between conflicts and conservation”. Final grade: 110/110 cum laude and Honorable Mention

Supervisor: Professor A. Bodini, co-supervisor: Dr. N. Clerici

### University of Ca’ Foscari, Venice

Sept. 2018 – Jul. 2021

Bachelor’s degree in Environmental Sciences

Thesis “Abiotic stresses mitigation by NK fertilizer on seeds of ancient cereals”. Final grade: 110/110 cum laude

Supervisor: Professor G. Buffa; co-supervisor: Dr. M. Roscini

## Work Experience

### Ecological Society of Australia

Dec. 2024 - Present

Lead Convenor of Special Interest Network "Integrated Forest Restoration"

### University of the Sunshine Coast

May 2025 – May 2026

High Degree by Research Student Representative on the Forest Research Institute (FRI)

Leadership Group

### University of the Sunshine Coast

Feb. 2024 – July 2024

Teacher assistant for “Discovering Science”

### University of Parma

Oct. 2022 – March 2023

Teacher assistant for “Biological laboratory techniques (Ecology module)”

## Selected publications

- **Vivian, J.**, Chazdon, R. L., Shapcott, A., & Lee, D. J. (2025). Toward a deeper understanding of recovery trajectories in forest ecosystem restoration: a machine learning approach. *Restoration Ecology*, e70483
- **Vivian, J.**, Chazdon, R. L., Catling, A. A., Shapcott, A., Herbohn, J., & Lee, D. J. (2026). *Acacia mangium* monocultures can catalyse the recovery of the tree community and aboveground carbon stock in the Philippines. *Forest Ecology and Management*, 601, 123357
- **Vivian, J.**, Scriver, M., Zaiko, A., Shapcott, A., Chazdon, R. L., Catling, A. A., & Lee, D. J. (2026). Recovery of soil microbiota in naturally regenerating *Acacia mangium* ecosystems. *PeerJ* 14:e21048
- **Vivian, J.**, Chazdon, R. L., Catling, A. A., & Lee, D. J. (2025). Early evidences of links between soil microbes and forest restoration through multiple ecosystem metrics. *Frontiers in Forests and Global Change*, 8, 1540513
- **Vivian, J.**, Shapcott, A., Chazdon, R. L., & Lee, D. J. (2026). Microbial rhizosphere and topsoil communities in young reforestation plantations. *Frontiers in Microbiology*, 17, 1915742.
- Catling, A. A., Derisso, V. D., Gamalo, L. E. D., Herbohn, J., McRae, C. L., Mooney, M. G., **Vivian, J.**, & Putz, F. E. (2026). Logging legacies in the Australian tropics: 56-years of skid trail recovery. *Forest Ecology and Management*, 619, 124075
- Bodini, A., **Vivian, J.**, Vargas, J., Clerici, N., & Scotti, M. (2024). The web of conflict-related interactions in Colombia: exploring causal linkages between ecological and social variables by the qualitative loop analysis. *Philosophical Transactions B*, 379(1909), 20230165

## Interests and Competences

|                                 |                                                                                                          |
|---------------------------------|----------------------------------------------------------------------------------------------------------|
| Languages                       | Italian (Mother tongue); English (TOEFL iBT, with a score of 108/120, corresponding to a C1; 22-10-2022) |
| Programming and software skills | R, QGIS, Python, LaTeX, Past 4.03, basics of “Weka”                                                      |
| Driving licenses                | A2-B (Italy); C-RE (Australia)                                                                           |

## Awards and Honors

|                                                                                                                                                                                                                                                                    |             |            |
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| • Grant “Quantum and Advanced Technologies Talent Building Program” for the development of research on merging quantum computing in ecology, from the Queensland Government                                                                                        | 20,000 AUD  | Oct. 2025  |
| • Culture Champion Award for “outstanding contributions helping to build and nurture the caring, supportive culture in the workplace”                                                                                                                              | 50 AUD      | Aug. 2025  |
| • Prize for the development of my research (from 1 <sup>st</sup> place in the 3-minute thesis competition at Forest Research Institute of the University of the Sunshine Coast                                                                                     | 500 AUD     | Jun. 2025  |
| • Travel bursary for Australian & New Zealand Environmental DNA (eDNA) Conference (Wellington)                                                                                                                                                                     | 2300 AUD    | Feb. 2025  |
| • Partial travel funding for Society of Ecological Restoration conference (Denver)                                                                                                                                                                                 | 200 USD     | Oct. 2025  |
| • Travel Award from the Ecological Society of Australia, for ESA conference (Melbourne)                                                                                                                                                                            | 150 AUD     | Dec. 2024  |
| • Travel Award from the Theo Murphy Initiative (Australia) administered by the Australian Academy of Science, Ecological Society of Australia, Plant-Soil Ecology Special Interest Network, to attend the ESA conference and workshops (Melbourne)                 | 1000 AUD    | Dec. 2024  |
| • Scholarship for Master’s degree (2023) final grade, from the local authority                                                                                                                                                                                     | 500 EUR     | Jan. 2024  |
| • Grant to participate in the Blended Intensive Programme “Towards a Sustainable Water Management” at the University of Extremadura, financed by ERASMUS+ SMS Short Mobility Grant, managed by the University of Parma in accordance with ERASMUS+ National Agency | Total cover | Jun. 2023  |
| • Grant to participate in the “International Summer School-Pollutions in Coastal Areas” at the University of Gdansk, from the project itself, financed under the Knowledge Education Development Operational Program of the EU Commission                          | Total cover | Sept. 2022 |
| • Scholarship of merit, 2 <sup>nd</sup> place, year 2018-2019, from Ca’ Foscari University of Venice                                                                                                                                                               | 1000 EUR    | Mar. 2020  |

## Conferences and Presentations

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| • European conference on ecological restoration (SERE2025; Brest, August 2026): presentation of part of my PhD and side-project results, entitled “Assessment of forest restoration projects: efficiency of cutting-edge techniques and comprehensive analysis” and “Optimized spatial planning in forest restoration via quantum algorithms: a proof of concept for viable plans” | Brest, Aug. 2026        |
| • Forestry Australia Conference: presentation “Trajectories of forest ecosystem recovery with <i>Acacia mangium</i> in the Philippines: what made it a successful story”                                                                                                                                                                                                           | Adelaide, Oct. 2025     |
| • World conference on ecological restoration (SER2025): presentation results “Can <i>Acacia mangium</i> plantations develop properties of remnant forests? A Machine Learning perspective”                                                                                                                                                                                         | Denver, Sept. 2025      |
| • International Conference on Conservation Biology (ICCB): presentation “Recovery of tree communities and above-ground carbon stock in <i>Acacia mangium</i> reforestation sites”                                                                                                                                                                                                  | Brisbane, Jun. 2025     |
| • Australian & New Zealand Environmental DNA (eDNA) Conference: presentation “Evaluating soil fungi and bacteria in <i>Acacia mangium</i> plantations and remnant forests in the Philippines using environmental DNA”                                                                                                                                                              | Wellington, Feb. 2025   |
| • Ecological Society Conference (ESA): poster presentation of early results                                                                                                                                                                                                                                                                                                        | Melbourne, Dec. 2024    |
| • University of the Sunshine Coast Showcase: poster presentation “The effect of reforestation using <i>Acacia mangium</i> on above- and below-ground ecosystem properties”                                                                                                                                                                                                         | Sippy Downs, Sept. 2024 |
| • Airborne eDNA workshop (attended online): presentation “Air DNA analysis and Machine Learning: a promising collaboration”                                                                                                                                                                                                                                                        | Canberra, Jun. 2024     |
| • Climate Exp0 (Online): presentation “Sustainable agriculture approaches”                                                                                                                                                                                                                                                                                                         | May 2021                |

## Relevant Accomplishments

Admitted to the PhD Program in Ecology at the University of Zurich/ETH Zurich in 2023. No PhD positions were available at the time.